

Devansh Parapalli

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Professional Summary

AI engineer who takes agent systems from first prototype to enterprise production. Joined Cisco's Finance agent platform as its sole engineer, built it into a shared framework other tracks deploy on, and led the support organization's move to working with AI agents. Technical architect on enterprise GenAI bids for TCS's GenAI Center of Excellence.

Work Experience

Systems Engineer Jul 2025 — Present
Tata Consultancy Services Limited Hyderabad, India

- Client: Cisco** — Scaled the team to 8 engineers under my technical lead within 6 months; own requirements, architecture, and cross-functional alignment with support, FinOps, and audit.
 - Support automation:** Reduced triage from a full business day to under 30 seconds with multi-intent detection, tool-calling, MCP servers, Kafka event streaming, and auto-scaling runners; agents now handle 750+ cases a month at 95% autonomous closure (4% pending human verification, 1% escalated).
 - Agent framework:** Architected the agent framework on Google ADK with backward-compatible A2A protocol extensions and a codegen layer that lets business users stand up agents themselves; powers 6 tracks and 8 agents, onboarding a new track in 4–5 hours.
 - Finance automation:** Eliminated 2,300+ person-hours of manual work a month across SOX payroll audit (1,000+) and FinOps reporting (1,300+) with AI-enabled pipelines over Snowflake, Oracle, and MongoDB, plus a self-built DAG framework for quarterly executive fund-flow reporting on highly restricted data; delivered a quarter-close and a year-end close with zero rework.
 - Data reconciliation platform:** Designed and built a reconciliation engine that compiles plain-English steps into processing and matching operations, running 100+ workflows at 1M+ rows per side with type-aware normalization that eliminates false positives.
 - Audit validation:** Built an independent audit layer for a production document-processing pipeline, using a deliberately different methodology so agreement between the two is real evidence; catches a real error in ~1 of 30 documents the primary pipeline passed at a 1-in-1,000 false-flag rate, sampling 30% of global invoices daily. Calibrated with eval sets and human review; replaced an LLM-as-a-Judge prototype on speed and cost.
 - Reskilling:** Led the AI reskilling of ~70 application-support engineers and managers into AI-native workflows; 75% now run AI agents in their day-to-day work.
- Internal: GenAI Center of Excellence** — Led technical architecture and client demos for 14 enterprise RFPs across GenAI enablement, cloud migration, and legacy modernization; 7 won, totalling \$450M+ TCV.
 - Architected an on-premises vLLM + LiteLLM inference platform serving 7B–120B parameter models with gateway routing, PII detection, RBAC, and audit logging, removing any external API dependency; handed off to a platform team for production rollout.
 - Authored a 21-use-case GenAI adoption roadmap for a European semiconductor manufacturer on Vertex AI and Gemini Enterprise, including on-premises infrastructure that keeps confidential engineering data inside their data center.
 - Architected the migration of a global telecom equipment maker's manufacturing software to GKE across all production sites.

GenAI & Backend Development Intern Jun 2024 — Aug 2024
PolymathAI Nagpur, India

- Cut per-video multi-modal processing time 12x, from 3 minutes to 15 seconds, by designing a parallelized inference pipeline across horizontally scaled containers; processed 14,600+ videos in production.

Projects

AIKO: AI-powered Knowledge Organizer (aiko.parapalli.dev) Jun 2024 — Present

- Tripled Mean Reciprocal Rank (0.25 to 0.75) and lifted nDCG@5 from 0.5 to 0.7 on a 100K-chunk technical corpus with hybrid retrieval (semantic search + keyword fallback), tuned chunking, and embedding selection across multi-modal content.
- Built a custom orchestration framework for full control over prompt construction, retrieval, and tool-calling, replacing third-party abstractions with task-specific logic.

Education

Government College of Engineering, Nagpur Nagpur, India
B.Tech, Computer Science and Engineering Dec 2021 — Jun 2025
CGPA: 8.28/10.0

Technical Skills

AI & Agents: LLM Serving (vLLM, LiteLLM), Multimodal Pipelines, Agentic Systems, Multi-Agent (A2A), MCP, LLM Evaluation, MLOps

Retrieval: RAG, Hybrid Retrieval, Embeddings, Vector DBs (pgvector, FAISS, Pinecone)

Backend & Systems: System Design, Distributed Systems, Event-Driven Architecture (Kafka), Async Concurrency, FastAPI

Data: Data Pipelines, Snowflake, PostgreSQL, Oracle, MongoDB, SQL

Cloud & Security: GCP (Certified Architect), Docker, Kubernetes, Terraform, OpenShift

Languages: Python, TypeScript, C/C++, SQL, PL/SQL

Certifications & Achievements

- GCP Certifications:** Professional Cloud Architect (Dec 2025) | Cloud Digital Leader (Sep 2025) | Generative AI Leader (Sep 2025) | [Credly](#)
- Publication:** "AI-based Knowledge Organizer for diverse data formats" - IJTE, March 2025